

Lecture -23

↳ Solving first order first derivative constant eqⁿ. using linear algebra.

$$\text{eq} \quad \frac{du_1}{dt} = -u_1 + 2u_2$$

$$\frac{du_2}{dt} = u_1 - 2u_2$$

$$A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}$$

$$u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} -1-\lambda & 2 \\ 1 & -2-\lambda \end{bmatrix}$$

$$\frac{du}{dt} = Au$$

$$(\lambda+1)(\lambda+2) - 2 = 0$$

$$\lambda^2 + 3\lambda = 0$$

$$\boxed{\lambda = 0, -3}$$

$$\lambda(\lambda+3) = 0$$

$$A - \lambda I = \begin{bmatrix} -1 - \lambda & 2 \\ 1 & -2 - \lambda \end{bmatrix}$$

$$x_1 = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 2 \\ 0 & 0 \end{bmatrix} \quad -x_1 + 2 = 0$$

$$\boxed{x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \quad \lambda_1 = 0$$

$$x_2 \rightarrow \begin{bmatrix} -1 - (-3) & 2 \\ 1 & -2 - (-3) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\boxed{x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}} \quad \lambda_2 = -3$$

$$U(t) = c_1 \underbrace{e^{\lambda_1 t}}_{\downarrow} x_1 + c_2 \underbrace{e^{\lambda_2 t}}_{\downarrow} x_2$$

Two solutions to the differential equation

Check : $\frac{du}{dt} = Au$

$$u = e^{\lambda_1 t} x_1$$

$$\frac{du}{dt} = \lambda_1 e^{\lambda_1 t} x_1$$

$$Au = A e^{\lambda_1 t} x_1$$

$$\Rightarrow u(t) = c_1 (e^{0t}) \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$u_t = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$u_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$u_0 = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 2c_1 - c_2 &= 1 \\ c_1 + c_2 &= 0 \end{aligned}$$

$$\underline{\underline{c_1 = 1/3}}$$

$$\underline{\underline{c_2 = -1/3}}$$

$$u_t = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \frac{1}{3} e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u_t = \frac{1}{3} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} - e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right)$$

As $t \rightarrow \infty$

steady state $u_\infty = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$

When do we get stability?

$\Rightarrow$ When the eigenvalues ≤ 0

OR

$\text{Real}(e^{\lambda t}) \rightarrow 0$ [more general as it includes complex numbers]

When do we have a steady state?

one eigenvalue is 0 and the others are negative real parts.

⇒ The solutions blow up when if any $\text{Real}(\lambda) > 0$

Note

Stability for 2×2 matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{trace}(A) = a + d = \lambda_1 + \lambda_2 < 0$$

$$\det A = ad - bc = \lambda_1 \lambda_2 > 0$$

⇒ This works for complex numbers too as they appear as complex conjugates.

$$\Rightarrow \frac{du}{dt} = Au$$

$$\text{Set } U = Sv$$

↓
decouple

$$S \frac{dv}{dt} = ASv$$

$$\frac{dv}{dt} = S^{-1}ASv = \Lambda v$$

$$\boxed{\frac{dv}{dt} = \Lambda v}$$

$$\frac{dv_1}{dt} = \lambda_1 v_1$$

⋮

#

$$\boxed{e^{At} = S e^{\Lambda t} S^{-1}}$$

wtf

$$e^{At}$$

$$\Rightarrow u_t = e^{At} u_0$$

$\Rightarrow$ Exponential of a matrix

$\rightarrow$ Use the power series of e^x (Taylor)

$$\Rightarrow e^{At} = I + At + \frac{1}{2} (At)^2 + \frac{(At)^3}{6} + \frac{(At)^n}{n!} \dots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{beautiful}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{GP} \quad \text{beautiful as well}$$

We can also use this Taylor GP series to find inverse of a matrix

$$(I - At)^{-1} = I + At + (At)^2 + (At)^3 \dots$$

→ If t is small $t^2, t^3 \dots$ ignored

$$\# e^{At} = S e^{\Lambda t} S^{-1}$$

$$e^{At} = I + At + \frac{(At)^2}{2} \dots$$

$$e^{At} = I + t(S\Lambda S^{-1}) + \frac{t^2(S\Lambda^2 S^{-1})}{2} \dots$$

$$e^{At} = S(\Lambda^0 S^{-1} + S(\Lambda t)^1 S^{-1} + \frac{S(\Lambda t)^2 S^{-1}}{2} \dots)$$

$$e^{At} = S \left(\Lambda^0 + (\Lambda t)^2 + \frac{(\Lambda t)^1}{2} \dots \right) S^{-1}$$

$$e^{At} = S e^{\Lambda t} S^{-1}$$

n independent eigenvectors

$$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & \ddots \\ & & & e^{\lambda_n t} \end{bmatrix}$$

eg second order differential $\rightarrow$

$$y'' + b y' + k y = 0$$

$$u = \begin{bmatrix} y' \\ y \end{bmatrix}$$

$$\boxed{y' = y'}$$

added

$$\Rightarrow u' = \begin{bmatrix} y'' \\ y' \end{bmatrix} = \begin{bmatrix} -b & -k \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y' \\ y \end{bmatrix}$$

$\Rightarrow$ Using this we can convert n^{th} order differential to $n \times n$ first order system.